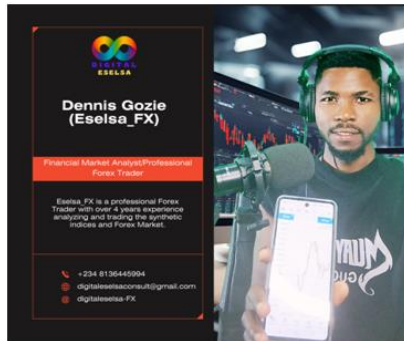




Practical Task: Secure File Management & User Configuration on Kali Linux

Objective:

Demonstrate your understanding of fundamental Linux file system and user management commands. You are tasked with setting up a secure workspace for a newly created user and organizing files with proper permissions and structure.

Instructions:

You **must not** use the root user to complete the file operations. A **new user must be created** and all steps (except `useradd` and `usermod`) must be performed using that user's account.

TASK STEPS:

1. Create a New User:

- Create a new user named `agent007` with a home directory and assign `/bin/bash` as their shell.
- Add this user to the `sudo` group.

```
sudo useradd -m -s /bin/bash -G sudo agent007
```

2. Switch to the New User:

- Set a password for `agent007`.
- Switch to the new user and perform the rest of the steps using this account.

3. Setup Workspace:

- Inside `/home/agent007/`, create a directory structure as follows:

```
/home/agent007/mission/  
├─ intel/  
├─ ops/  
└─ backup/
```

- Use `mkdir` and `cd` commands to navigate and create this structure.
 - 4. File Creation and Listing:**
 - Inside `intel/`, create three empty files: `targets.txt`, `routes.txt`, and `keys.txt`.
 - Use `ls`, `ls -l`, and `ls -la` to list contents and view permissions of the files and directories. Redirect the output of each listing to separate files named `listing1.txt`, `listing2.txt`, and `listing3.txt` inside the `intel/` folder.
 - 5. Copy and Remove Files:**
 - Copy `routes.txt` to the `backup/` folder.
 - Delete `keys.txt` from `intel/` using `rm`.
 - 6. Permissions Handling:**
 - Change the permissions of `targets.txt` so that:
 - Owner can read and write.
 - Group can only read.
 - Others have no permissions.
 - Use `chmod` and verify using `ls -l`.
 - 7. Simulate an Error and Recover:**
 - Intentionally **remove the `ops/backup/routes.txt` file**.
 - Copy the `routes.txt` from `intel/` back into the `backup/` folder.
 - (This tests `cp` and basic error recovery.)
 - 8. Advanced Permissions & Ownership:**
 - As root, change the ownership of `intel/targets.txt` to another user (e.g., `root`), then switch back to `agent007` and observe the result using `ls -l`.
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✔ Submission Requirements:

Students must submit:

- The command history (`.bash_history` or a copy-paste of commands used).
- A screenshot or text file showing the final `ls -laR /home/agent007/mission/` output.
- A text explanation (4–5 lines) of what `chmod`, `ls -l`, and `useradd` do.